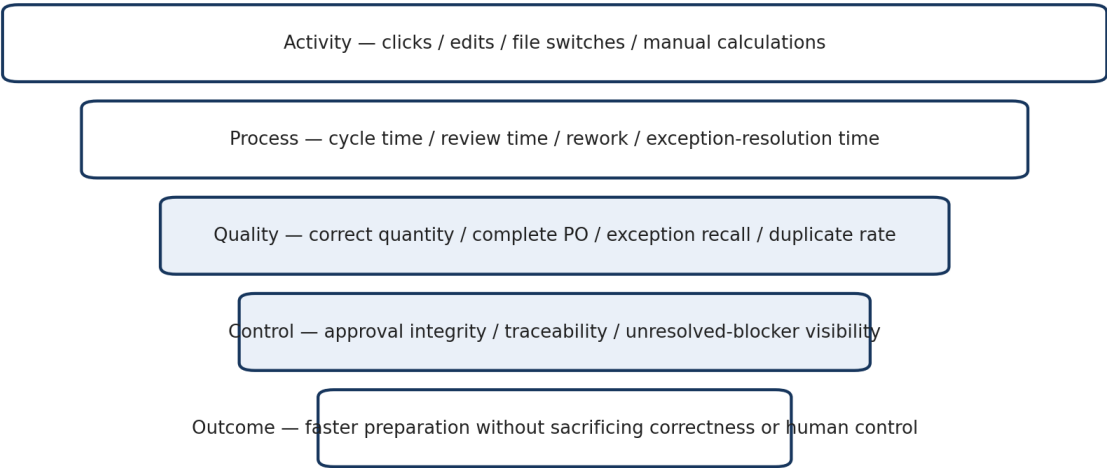


Measurement and Instrumentation Plan

Timing, user actions and quality measures

Purpose. Define what should be recorded during manual and agent-assisted runs so that performance can be measured rather than estimated.

What we measure - from activity to business outcome



1. Timestamp checkpoints

Checkpoint	Definition
T0	Task begins / input pack opened
T1	Inputs loaded and run started
T2	First complete decision set available
T3	Draft ready for human review
T4	Human review / final action completed

2. Core derived time measures

Metric	Formula	Interpretation
Preparation time	$T3 - T0$	How long until a reviewable draft exists.
System processing time	$T2 - T1$	Agent / tool execution rather than human work.
Review time	$T4 - T3$	Human effort after the agent has prepared output.
Total task time	$T4 - T0$	End-to-end user experience.
Exception discovery latency	time first blocker shown - $T0$	How early the workflow exposes a problem.

3. Action counts

Field	Counting rule
file_switch_count	Number of times user changes file / worksheet / source.
lookup_count	Manual searches for SKU / supplier / price / constraint.
calculation_count	Manual arithmetic / helper-cell calculation.
field_edit_count	Manual data entry or correction.
review_edit_count	Changes made after a draft is generated.
rework_cycle_count	Return to an earlier step because of an error / missing input.
raw_file_reopen_count	Times reviewer must return to source data to understand a line.

4. Quality measures

Metric	Definition
Decision accuracy	$\text{Correct reorder / no-reorder outcomes} \div \text{tested outcomes}$
Quantity accuracy	$\text{Correct final quantity outcomes} \div \text{quantity cases}$
Exception recall	$\text{Expected blocking exceptions shown} \div \text{expected blocking exceptions}$
False-block rate	$\text{Items incorrectly blocked} \div \text{valid items}$
PO completeness	$\text{Required populated PO fields} \div \text{required fields}$
Traceability completeness	$\text{PO lines with usable source reference} \div \text{PO lines}$
Approval integrity	$\text{Approval-control cases passed} \div \text{approval-control cases}$

Metric	Definition
Duplicate draft rate	Duplicate active drafts ÷ repeated-generation cases

5. Suggested event log schema

Field	Purpose
run_id	Identifies one procurement run.
scenario_id	Links test / experiment case.
timestamp	Event time.
actor	user / agent / system.
event_type	input_loaded, decision_created, exception_created, draft_created, edit, approved, held, etc.
sku	Optional SKU reference.
supplier_id	Optional supplier reference.
status_before / after	State transition.
field_changed	For review edits.
reason_code	Machine-readable reason / exception.
source_ref	Input / decision traceability pointer.

6. Reviewer pulse survey (30 seconds)

Item	Statement	Scale
Q1	I could tell why each line was included.	1-5
Q2	I could identify blocked items quickly.	1-5
Q3	I trusted the totals without recalculating everything.	1-5
Q4	I knew what still required my judgement.	1-5
Q5	I would prefer this workflow to the manual process for the same task.	1-5